

COMP 3069

Computer Graphics



Lecture 1:
Overview of the Module

Autumn 2025

Module Information

- ◆ Instructors: Prof Sean He,
Dr Wooi Ping Cheah
- ◆ Lecture Time:
 - Monday: 1pm - 3pm (DB-A06)
- ◆ Lab Session Time:
 - Wednesday: 9am - 12pm (IAMET-406)

Instructor's Contacts

◆ Prof. Sean He

- Email: Sean.He@nottingham.edu.cn
- Office: PMB 334
- Office Hours: *Monday 3-5pm (From 10 November - 15 December 2025)*
OR, send email to make appointment ...

Instructor's Contacts

◆ Dr. Wooi Ping Cheah

- Email: Wooi-Ping.Cheah@nottingham.edu.cn
- Office: PMB 323
- Office Hours: *Thursday 4pm - 6pm*

OR, send email to make appointment ...

Teaching Assistant

- ◆ *Teaching Assistant* (GTA) to help you around ...
 - *Ms. Yue Li*
 - Email: Yue.LI3@nottingham.edu.cn

Textbooks and References

- ◆ Textbook: (Primary reference)
 - **Interactive Computer Graphics: A Top-Down Approach with Shader-Based OpenGL**, 8th Edition, by *Edward Angel and Dave Shreiner*, Published by: *Pearson* in 2019
- ◆ Secondary reference:
 - **Computer Graphics with OpenGL**, 4th Edition, by *Donald Hearn, Pauline M. Baker, and Warren R. Carithers*, Published by: *Pearson* in 2014



Module Aims

- ◆ Understand the principles and techniques of 2D and 3D computer graphics
- ◆ Develop 2D and 3D computer graphics programming skills using OpenGL
- ◆ You will learn
 - Fundamental computer graphics principles and techniques
 - 2D and 3D graphics programming
 - To write cool graphics programs



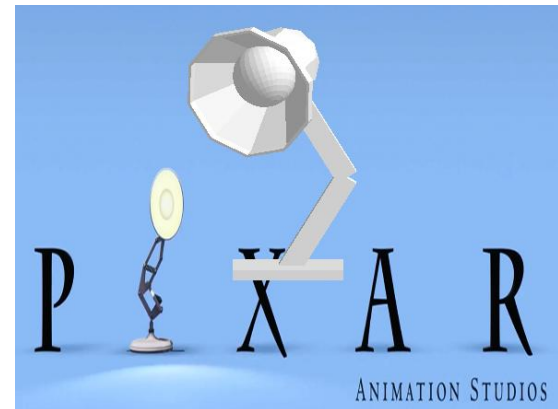
Lectures

- ◆ In the lectures, you learn computer graphics principles and techniques
- ◆ You learn how mathematics is used in graphics programs
- ◆ In principle,
 - This module is about "*Graphics*" on Computers ...
 - This module is not just "*Mathematics*" ...
 - This module is not just "*Programming*" ...
 - This module is :
@ *Graphics*+*Mathematics*+*Programming*



Labs

- ◆ You learn graphics programming by doing programming in the labs / home
- ◆ Lab sessions are designed to synchronize with the lectures
- ◆ The aim is to help you
 - understand the lectures better through practice
 - develop your coursework



Labs



- ◆ The labs use OpenGL and run on PC.
- ◆ OpenGL is the premier environment for developing 2D and 3D graphics applications.

C++ Programming

- ◆ Since you came here you probably want to learn the inner workings of computer graphics, there are a few items that need to be taken into consideration.
- ◆ Since OpenGL is a **graphics API** and not a platform of its own, it requires a language to operate in and the language of choice is **C++**, therefore **you will be doing some C++** programming, but we are NOT teaching C++.
- ◆ Free tutorials at www.learncpp.com



Some Maths Needed

- ◆ Also, we will be using some math (*linear algebra*, *geometry* and *trigonometry*) along the way. Do not be scared about the mathematical knowledge required
- ◆ Almost all the concepts can be understood with a basic mathematical background, and we will try to keep the mathematics to a minimum where possible.
- ◆ Most of the functionality does not even require you to understand all the math as long as you know how to use it.



Teaching Materials

- ◆ Lecture notes will be available on Moodle.
- ◆ A lot of the classic books are still valid:
 - “*Computer Graphics: Principles and Practice*” by Hughes et al.
 - “*Real Time Rendering*” by Moller/Haines
 - “*Physically Based Rendering*” by Pharr
 - “*Game Engine Architecture*” by Gregory
 - “*Mathematics for 3D Game Programming and Computer Graphics*” by Lengyel

Learning more ...

- ◆ Geometry, illumination, shaders and tools etc. will take a lot of time to master.
- ◆ Start with modern OpenGL, and work through computer graphics theory (and a lot of practice) using those.
- ◆ Online resources:
<https://learnopengl.com/>
<http://graphicscodex.com/>

...



Teaching Plan (Provisional) 2025-2026

	Week beginning	Lecture Mondays, 1pm -- 3pm (DB-A06)	Lab Wednesdays, 9am -- 12pm (IAMET-406)
Week 1	15 Sep	No lecture (Induction Week)	No lab (Induction Week)
Week 2	22 Sep	Overview of the Module (Wooi Ping Cheah)	No lab
Week 3	29 Sep	Rendering (Wooi Ping Cheah)	No lab (National Day)
Week 4	6 Oct	No lecture (National Day)	Rendering (Wooi Ping Cheah)
Week 5	13 Oct	Graphics Programming OpenGL & GLSL (Wooi Ping Cheah)	Graphics Programming (Wooi Ping Cheah)
Week 6	20 Oct	Mathematics for Graphics (Wooi Ping Cheah)	2D Modelling (Wooi Ping Cheah)
Week 7	27 Oct	Transformations (Wooi Ping Cheah)	2D Transformations (Wooi Ping Cheah)
Week 8	3 Nov	3D to 2D & Spaces (Wooi Ping Cheah)	3D Modelling (Wooi Ping Cheah)
Week 9	10 Nov	Rendering Pipeline (Sean He)	Rendering Pipeline (Sean He)
Week 10	17 Nov	Camera (Sean He)	Camera (Sean He)
Week 11	24 Nov	Textures (Sean He)	Textures (Sean He)
Week 12	1 Dec	Light (Sean He)	Light (Sean He)
Week 13	8 Dec	Antialiasing (Sean He)	Antialiasing (Sean He)
Week 14	15 Dec	No lecture (Self-Study)	No lab (Self-Study)



Assessment Methods

Assessment	Weight (%)	Duration	Requirements
In-class Quiz (Week 8)	15	30 minutes	The students will answer questions to demonstrate that they understand computer graphics principles.
In-class Lab Test (Week 10)	15	1.5 hours	The students will program 2D graphics to demonstrate that they understand computer graphics techniques.
Coursework	70	2 weeks	Programming project designing and implementing a 3D graphics scene, plus a written report with a demonstration of the project.

The Coursework/Project

- ◆ Your knowledge and creativity count - **you decide** on a graphical environment or application and implement it for your coursework
- ◆ You apply the techniques taught and/or what you have learnt yourself - we encourage **self-learning**



What former CG students say

- ◆ Learning and understanding how a sphere can be built using this technique was very interesting and I realised how **3D graphical modelling and animation is all about the clever use of trigonometry, geometry and mathematics.**
- ◆ I believe all of the techniques I used to develop this coursework were covered in the lectures, labs and examples provided to us. The framework was an incredibly useful tool which made it easy for me, as a **Java developer**, to understand the underlying **object oriented principles** used in Computer Graphics.



Module Content

- ◆ Introduction
- ◆ Shading, Rasterisation
- ◆ Graphics APIs, 2D Geometry
- ◆ Mathematics
- ◆ Transformations and Matrices
- ◆ 3D Geometry
- ◆ Rendering Pipeline
- ◆ Cameras
- ◆ Texture Mapping
- ◆ Lighting Models
- ◆ Antialiasing
- ◆ ...



Computer Graphics

- ◆ 3D computer graphics is the discipline of modelling the real world, e.g., creating 3D objects/scenes, and turning them into realistic 2D images on the computer screen
- ◆ The process is analogous to taking a photograph of a 3D scene or object, using a digital camera
- ◆ Computer graphics has a wide range of applications, including games and entertainment, media production, computer-aided design, virtual/augmented/mixed reality for engineering, medicine, and education, etc.



Application: Entertainment - Games, Films, TV, Media Production



Application: Science & Engineering - Visualisation, CAD, Virtual Prototyping



Application: Education & Training

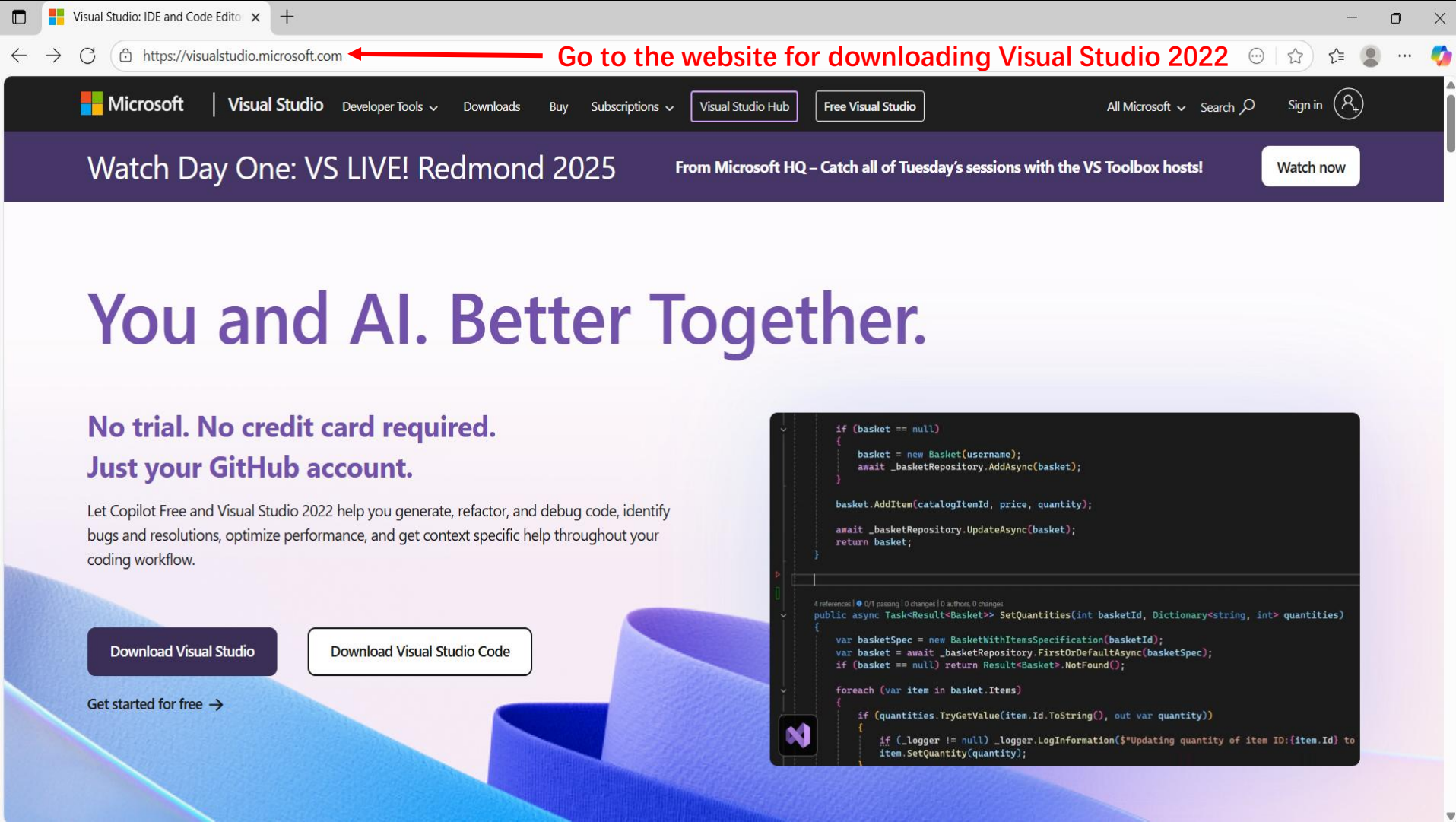


Installing Microsoft Visual Studio 2022

Microsoft Visual Studio 2022 is the latest version of the software. It is available on all lab PCs for your coursework. While students are strongly encouraged to utilise the lab facilities, they are also permitted to use personal computers or laptops for their work.

Our first lab session is on Wednesday, 8 October 2025 (Week 4). To ensure a smooth session and avoid network congestion from simultaneous downloads, please install the software on your personal machines before coming to the lab session.

The following slides provide a step-by-step guide for downloading and installing Visual Studio 2022. The subsequent setup and configuration of OpenGL and all necessary tools for this course will be guided during the lab session.



Visual Studio: IDE and Code Editor x +

← → ↻ <https://visualstudio.microsoft.com> ← **Go to the website for downloading Visual Studio 2022** ⋮ ⭐ ⚙️

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Watch Day One: VS LIVE! Redmond 2025 From Microsoft HQ – Catch all of Tuesday's sessions with the VS Toolbox hosts! Watch now

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Just your GitHub account.

Let Copilot Free and Visual Studio 2022 help you generate, refactor, and debug code, identify bugs and resolutions, optimize performance, and get context specific help throughout your coding workflow.

[Download Visual Studio](#) [Download Visual Studio Code](#)

[Get started for free →](#)

```
if (basket == null)
{
    basket = new Basket(username);
    await _basketRepository.AddAsync(basket);
}

basket.AddItem(catalogItemId, price, quantity);

await _basketRepository.UpdateAsync(basket);
return basket;
}

4 references | 0/1 passing | 0 changes | 0 authors, 0 changes
public async Task<Result<Basket>> SetQuantities(int basketId, Dictionary<string, int> quantities)
{
    var basketSpec = new BasketWithItemsSpecification(basketId);
    var basket = await _basketRepository.FirstOrDefaultAsync(basketSpec);
    if (basket == null) return Result<Basket>.NotFound();

    foreach (var item in basket.Items)
    {
        if (quantities.TryGetValue(item.Id.ToString(), out var quantity))
        {
            if (_logger != null) _logger.LogInformation($"Updating quantity of item ID:{item.Id} to {quantity}");
            item.SetQuantity(quantity);
        }
    }
}
```

Visual Studio: IDE and Code Editor x +

https://visualstudio.microsoft.com

Pick the right tool for you

Visual Studio with AI is built to enhance your productivity at every stage of development.



Visual Studio | Windows

The most comprehensive IDE for .NET and C++ developers on Windows. Fully packed with a sweet array of tools and features to elevate and enhance every stage of software development.

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[Download Visual Studio](#) ▾

- Community 2022
- Professional 2022
- Enterprise 2022



Visual Studio Code | Windows, macOS, Linux

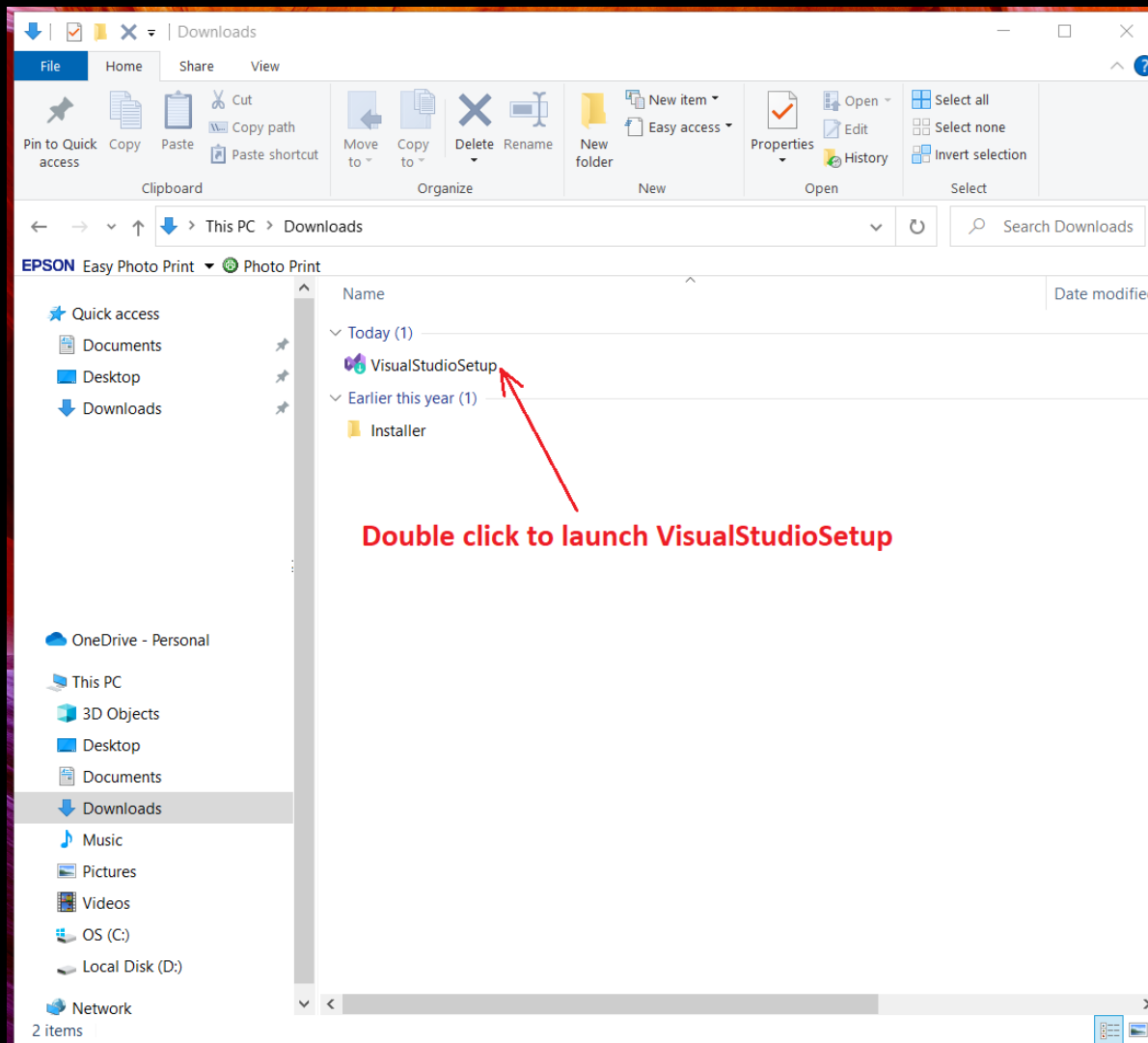
A standalone source code editor that runs on Windows, macOS, and Linux. The top pick for JavaScript and web developers, with extensions to support just about any programming language.

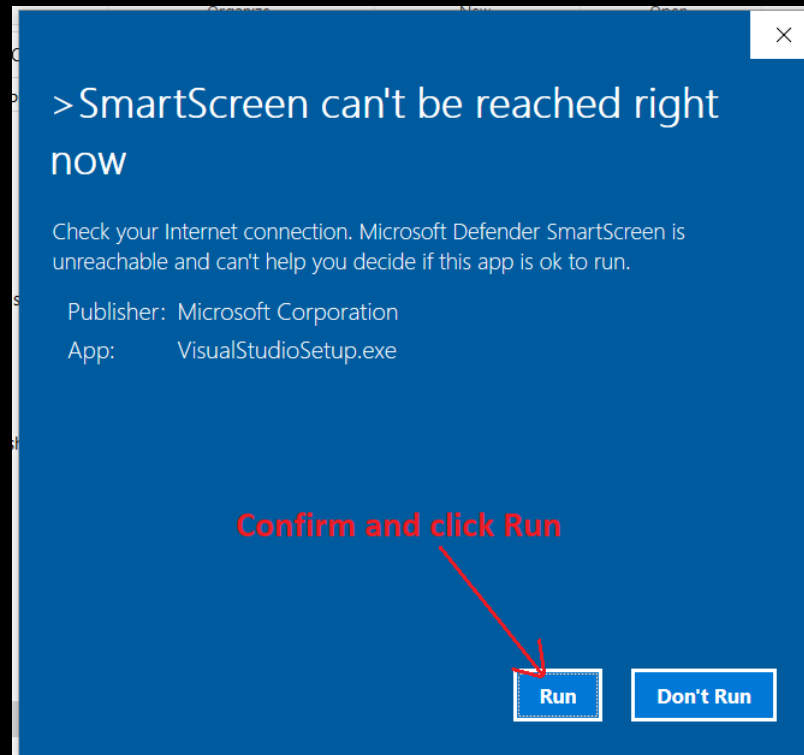
[Learn more →](#)

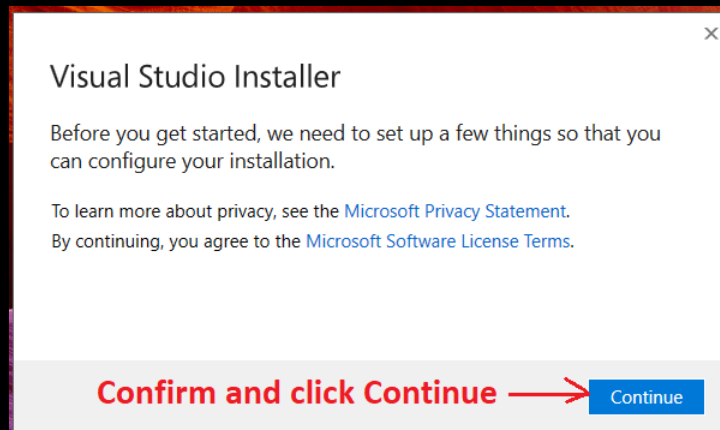
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[Download Visual Studio Code](#) ▾

<https://visualstudio.microsoft.com/thank-you-downloading-visual-studio/?sku=Community&channel=Release&version=VS2022&source=VSLandingPage&cid=2030&passive=false>







Visual Studio Installer

Getting the Visual Studio Installer ready.

Downloading: 4.85 MB of 21.92 MB

1.58 MB/sec

Installing

Take a few minutes to download and install

Cancel

Visual Studio Installer

 Getting the Visual Studio Installer ready.



Download complete

Installing — Visual Studio Community 2022 — 17.6.2

Workloads Individual components Language packs Installation locations

Need help choosing what to install? [More info](#)

Web & Cloud (4)

Choose what to install



ASP.NET and web development



Build web applications using ASP.NET Core, ASP.NET, HTML/JavaScript, and Containers including Docker supp...



Azure development



Azure SDKs, tools, and projects for developing cloud apps and creating resources using .NET and .NET Framework...

Scroll down



Python development



Editing, debugging, interactive development and source control for Python.



Node.js development



Build scalable network applications using Node.js, an asynchronous event-driven JavaScript runtime.

Desktop & Mobile (5)



.NET Multi-platform App UI development



Build Android, iOS, Windows, and Mac apps from a single codebase using C# with .NET MAUI.



.NET desktop development



Build WPF, Windows Forms, and console applications using C#, Visual Basic, and F# with .NET and .NET Frame...

Location

C:\Program Files\Microsoft Visual Studio\2022\Community [Change...](#)

Remove out-of-support components

By continuing, you agree to the [license](#) for the Visual Studio edition you selected. We also offer the ability to download other software with Visual Studio. This software is licensed separately, as set out in the [3rd Party Notices](#) or in its accompanying license. By continuing, you also agree to those licenses.

Total space required 1.17 GB

Install while downloading

Install

Installing — Visual Studio Community 2022 — 17.6.2

Workloads

Individual components

Language packs

Installation locations

Need help choosing what to install? [More info](#)



Desktop development with C++

Build modern C++ apps for Windows using tools of your choice, including MSVC, Clang, CMake, or MSBuild.



Universal Windows Platform development

Create applications for the Universal Windows Platform with C#, VB, or optionally C++.



Mobile development with C++

Build cross-platform applications for iOS, Android or Windows using C++.



Gaming (2)



Game development with Unity

Create 2D and 3D games with Unity, a powerful cross-platform development environment.



Game development with C++

Use the full power of C++ to build professional games powered by DirectX, Unreal, or Cocos2d.



Other Toolsets (5)

Location

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By continuing, you agree to the [license](#) for the Visual Studio edition you selected. We also offer the ability to download other software with Visual Studio. This software is licensed separately, as set out in the [3rd Party Notices](#) or in its accompanying license. By continuing, you also agree to those licenses.

Installation details

Visual Studio core editor

Desktop development with C++

Included

✓ C++ core desktop features

Optional

✓ MSVC v143 - VS 2022 C++ x64/x86 build t...

✓ C++ ATL for latest v143 build tools (x86 &...

✓ Windows 11 SDK (10.0.22000.0)

✓ Security Issue Analysis

✓ Just-In-Time debugger

✓ C++ profiling tools

✓ C++ CMake tools for Windows

✓ Test Adapter for Boost.Test

✓ Test Adapter for Google Test

✓ Live Share

✓ IntelliCode

✓ C++ AddressSanitizer

✓ vcpkg package manager

□ C++ MFC for latest v143 build tools (x86...

Remove out-of-support components

Total space required 9.06 GB

Install while downloading

Install

Choose "Desktop development with C++" to install

Click Install

Visual Studio Installer

**Install Visual Studio 2022 Community
with the selected components**

Installed Available



Visual Studio Community 2022

Pause

Downloading and verifying: 187 MB of 2.44 GB

(7 MB/sec)

7%

Installing: package 12 of 440

0%

Microsoft.VisualStudio.Platform.Editor

☒ Start after installation

[Release notes](#)

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Monday, June 12, 2023

[Visual Studio's IntelliSense list can now steer
GitHub Copilot code completions.](#)

TL:DR GitHub Copilot and Visual Studio's built-in...

Thursday, June 8, 2023

[Announcing C# Dev Kit for Visual Studio Code](#)

We are thrilled to announce the preview release o...

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Need help? Check out the [Microsoft Developer
Community](#) or reach us via [Visual Studio
Support](#).

Installer version 3.6.2115.31769

Visual Studio Installer

Installation complete

Click to end installation,
and reboot system

Installed Available

All installations are up to date.



Visual Studio Community 2022

17.6.2

Powerful IDE, free for students, open-source contributors, and individuals

[Release notes](#)

Modify

Launch

Done installing

Visual Studio has been successfully installed. We recommend rebooting soon to clean up any remaining files.

OK

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